



An adaptive testing item selection strategy via a deep reinforcement learning approach

Pujue Wang^{1,2,3} · Hongyun Liu^{2,3} · Mingqi Xu⁴

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Abstract

Computerized adaptive testing (CAT) aims to present items that statistically optimize the assessment process by considering the examinee's responses and estimated trait levels. Recent developments in reinforcement learning and deep neural networks provide CAT with the potential to select items that utilize more information across all the items on the remaining tests, rather than just focusing on the next several items to be selected. In this study, we reformulate CAT under the reinforcement learning framework and propose a new item selection strategy based on the deep Q-network (DQN) method. Through simulated and empirical studies, we demonstrate how to monitor the training process to obtain the optimal Q-networks, and we compare the accuracy of the DQN-based item selection strategy with that of five traditional strategies—maximum Fisher information, Fisher information weighted by likelihood, Kullback–Leibler information weighted by likelihood, maximum posterior weighted information, and maximum expected information—on both simulated and real item banks and responses. We further investigate how sample size and the distribution of the trait levels of the examinees used in training affect DQN performance. The results show that DQN achieves lower RMSE and MAE values than traditional strategies under simulated and real banks and responses in most conditions. Suggestions for the use of DQN-based strategies are provided, as well as their code.

Keywords Computerized adaptive testing · Item selection strategy · Reinforcement learning · Deep learning · Deep Q-network

Introduction

Computerized adaptive testing (CAT) is a widely used tool of smart testing, as well as a popular research topic in psychological and educational measurement (Zhang & Chang,

2016). In comparison to nonadaptive test formats such as paper-and-pencil tests, CAT can use fewer items to offer a more accurate estimate of the examinee's true trait levels. This is achieved by presenting a group of the most statistically informative items, considering both the examinee's responses and constraints such as content balancing and exposure control (Weiss, 1982; Wainer et al., 2000; Frey, 2023). Item selection, as the core process used to achieve efficient measurement, has been an important research focus in CAT. Traditional item selection strategies that serve as applications are mostly proposed under the framework of item response theory (IRT; van der Linden, 2016). The most widely used method involves choosing items that maximize the Fisher information (MFI; Lord, 1980). Within the context of IRT, the Fisher information statistic is inversely related to the asymptotic variance of the maximum likelihood estimator of the trait level. Thus, the MFI strategy obtains a small standard error by simply choosing the item with maximal Fisher information at the current trait level.

✉ Hongyun Liu
hlyiu@bnu.edu.cn

¹ Beijing Key Laboratory of Learning and Cognition, School of Psychology, Capital Normal University, No. 23 Bai Dui Zi Jia, Beijing 100048, China

² Beijing Key Laboratory of Applied Experimental Psychology, National Demonstration Center for Experimental Psychology Education (Beijing Normal University), Faculty of Psychology, Beijing Normal University, No. 19 Xin Jie Kou Wai Street, Beijing 100875, China

³ Faculty of Psychology, Beijing Normal University, No. 19 Xin Jie Kou Wai Street, Beijing 100875, China

⁴ Department of Psychology, Tufts University, 419 Boston Avenue, Medford, Massachusetts 02155, USA

The item selection strategy that maximizes Fisher information is not a perfect tool. One of the major disadvantages of this popular method is that Fisher information is not the appropriate indicator when the estimated trait level is far from the true trait level (Chang & Ying, 1996). Therefore, MFI may not always select the most informative item given an inaccurate trait estimator, especially in the early stage of the test (Chen et al., 2000). On the other hand, the distribution of the estimated trait level will gradually converge as the test proceeds, but this updated distribution is not considered in selecting the item (Samejima, 1977). The problem of MFI has inspired abundant research on improvements in strategies. To address the disadvantages above, some alternative strategies have been proposed to improve efficiency from different perspectives. Within the framework of IRT, researchers have improved a variety of item selection strategies focusing on the item information index (see Section "CAT item selection strategies based on IRT" for details).

As new technologies and tools have been introduced in smart education in recent years, new directions for improving classical information-related item selection strategies have emerged. First, most traditional IRT-based strategies make local optimal choices rather than globally optimal ones. Despite the multiple variants of item information, most strategies consider only the information pertaining to the next item. It remains difficult for classical strategies to take into account the potential information gained by considering a few more items. Although the shadow test (van der Linden & Reese, 1998) aims to plan for all remaining test items, each item is still selected based on the current estimated trait level to maximize the information for just the next item. If the trait level changes significantly later, these early locally optimal choices may be suboptimal for the entire test. There may be no items with high information for the estimated trait levels in the middle and late stages if the global optimal selection path is not planned in the early stage. The ideal strategy should be forward-looking. Each item should be selected to maximize the information that can be obtained in the subsequent test, while considering the variations in the estimated trait levels and the item selection sequences that may occur later. Second, traditional strategies do not consider mining information from other examinees' testing records to improve testing efficiency. IRT-based strategies can only utilize large-scale data to update the item parameters in the IRT model and cannot directly learn from the experience of other examinees due to the theoretical framework. By learning from the previous testing records, the ideal selection strategy based on a flexible framework may predict the possible trajectories of change in the trait estimation and, more importantly, the test information obtained in the future after selecting any item.

A promising approach to addressing these shortcomings is reinforcement learning (RL). As an interdisciplinary field

inspired by behavioral psychology, RL is adept at training an intelligent agent to explore superior strategies for performing complicated tasks (Sutton & Barto, 2018). As deep learning technology is rapidly developing, a group of innovative algorithms combining RL and neural networks, referred to as "deep reinforcement learning" (DRL), have surpassed traditional methods and even human intelligence in various frontier domains, such as AlphaGo (Silver et al., 2016, 2017). In recent years, RL has been preliminarily proven to be suitable and powerful for solving problems with long-term and adaptive objectives in education measurement, such as learning material recommendation in adaptive learning systems (Chen et al., 2018; Han et al., 2020; Li et al., 2021, 2023; Tan et al., 2020; Tang et al., 2019). The similarity between adaptive learning and adaptive testing, namely, that they both aim to make personalized recommendations based on individual characteristics and to reduce the number of materials or items to enhance efficiency, also provides the basis for introducing RL into CAT. In fact, a few pioneering studies have adopted RL frameworks and DRL algorithms in different types of CAT to develop selection strategies (Nurakhmetov, 2019; Ghosh & Lan, 2021; Shin & Bulut, 2022).

In the classification CAT, Nurakhmetov (2019) redefined the elements of CAT under the RL framework and proposed a sequential selection strategy based on the DRL algorithm. This algorithm achieved classification accuracy of 89.8% with just 10 items, surpassing the baseline strategy's 85.5% accuracy with 15 items. Furthermore, the accuracy increased to 92.1% with 15 items using the DRL-based approach. Shin and Bulut (2022) adopted the RL framework and algorithm introduced by Nurakhmetov (2019) in computerized formative assessments to select the optimal test to administer to students based on their learning trajectories. Relatively high accuracy (close to 90%) was achieved in significantly fewer test administrations using this recommendation system, as indicated by F1 scores above 0.90. Item selection strategies under RL frameworks have also been proposed for large-scale online education systems. Ghosh and Lan (2021) constructed a bilevel optimization-based framework for CAT to simultaneously model examinees and learn selection strategies from real data. The new framework had a lower trait estimation error and reached the same accuracy while using up to 30% fewer items than the method that matches the difficulty parameter under the Rasch model.

In summary, RL provides CAT with the possibility of considering both current and long-term information gains in each item selection, thus compensating for the short-sightedness of traditional strategies that only consider the information of the current item. Therefore, using reinforcement learning as the framework to design a selection strategy can yield high measurement accuracy with fewer items. Moreover, DRL can accurately find the optimal path in complex situations to achieve an adaptive goal without exploring all

possible paths. As each choice follows the global optimal path, the item selection strategy based on DRL may further reduce the measurement error with fewer items in complex CAT scenarios.

The following gaps remain in existing research on the incorporation of RL into CAT. First, no RL framework has been defined and no selection strategy has been proposed for CAT based on the most common IRT models, such as the three-parameter logistic model (3PLM). The RL framework used in previous studies is not suitable for common IRT models. In Nurakhmetov's (2019) setting, which was also used by Ghosh and Lan (2021), the latent traits of examinees were represented as latent classes rather than continuums. To accord with this framework, more complex recurrent neural networks were used to estimate the latent class instead of reducing the computational complexity by means of mature psychometric models. Second, the DRL algorithm most commonly used by existing studies to implement the selection strategy, the actor–critic algorithm, has to train two neural networks and often requires a substantial amount of training data. In the studies mentioned above, the number of examinees ranged from a thousand to hundreds of thousands. Third, there is a lack of research that assesses whether the RL-based item selection strategy designed for the commonly used IRT model can improve the measurement accuracy by accounting for long-term information as well as the factors that can affect the performance of the new strategy. In addition, no comparison between the new RL-based strategy and classical IRT-based CAT selection strategies has been made under the most common CAT scenario described above.

To address these issues, this paper aims to redefine the item selection problem using a classical and fundamental RL framework, the Markov decision process (MDP), for the most common unidimensional dichotomous CAT. On this basis, some IRT concepts and methods are incorporated into the MDP framework to reduce the difficulty of deployment and implementation and enhance comparability with traditional strategies. Additionally, a simple and efficient DRL algorithm, the deep Q-network (DQN; Mnih et al., 2013, 2015), is used to implement the item selection strategy. As the representative algorithm of RL that combines deep neural networks, DQN has been successfully applied in many fields, including video games (Mnih et al., 2013, 2015), robotics (Gu et al., 2017), and autonomous driving (Sallab et al., 2017). A study by Li et al. (2023) in the field of adaptive learning has shown that combining DQN with models that simulate examinees' transfer states can significantly reduce the amount of training data needed.

The remainder of the article is organized as follows. Section "Deep Q-network for CAT item selection" gives the fundamental concepts of the MDP framework and its corresponding elements in the CAT scenario and then proposes a new item selection strategy based on the DQN algorithm.

Sections "Simulation study" and "Empirical study based on the real bank and responses" present two studies and corresponding results, which compare the accuracy of the new strategy and five traditional IRT-related strategies in different situations. Section "Discussion" concludes with a summary of the findings and a discussion of the use of DQN in CAT.

Deep Q-network for CAT item selection

CAT item selection strategies based on IRT

Taking 3PLM as an example, the probability $P_i(\theta)$ of giving a correct response to item i is as follows:

$$P_i(\theta) = c_i + (1 - c_i) \frac{\exp[a_i(\theta - b_i)]}{1 + \exp[a_i(\theta - b_i)]} \quad (1)$$

where θ is the trait level,¹ and a_i , b_i , and c_i correspond to the discrimination, difficulty, and pseudo-guessing parameters for item i , respectively.

The Fisher information function is computed as

$$I_i(\theta) = \frac{[P'_i(\theta)]^2}{P_i(\theta)[1 - P_i(\theta)]} = \frac{a_i^2 (1 - c_i)}{[c_i + \exp[a_i(\theta - b_i)]] [1 + \exp[-a_i(\theta - b_i)]]^2} \quad (2)$$

where $P'_i(\theta)$ is the first-order derivative of $P_i(\theta)$ with respect to θ . The item number for the l th item in CAT is denoted as i_l , and the rule of MFI is described as

$$i_l = \underset{i \in B_l}{\operatorname{argmax}} I_i(\hat{\theta}_l) \quad (3)$$

where $\hat{\theta}_l$ is the trait level estimated by maximum likelihood estimation (MLE), and B_l represents the subset of items that have not been administered when selecting the l th item.

Veerkamp and Berger (1997) proposed a new strategy that maximizes the Fisher information weighted by the likelihood over all possible trait levels (abbreviated as FIWL), described as

$$i_l = \underset{i \in B_l}{\operatorname{argmax}} \int_{-\infty}^{\infty} I_i(\hat{\theta}_l) L(\hat{\theta}_l) d\theta \quad (4)$$

where $L(\theta)$ is the likelihood function.

The Kullback–Leibler information (KL; Chang & Ying, 1996), also referred to as the global measure of information, evaluates the item discrimination capacity by considering a broader range of trait levels. The KL information-based

¹ For simplicity, the subscript that indicates the specific examinee is omitted. It is added only when it is necessary to differentiate trait levels across distinct individuals.

selection strategy is commonly weighted by the likelihood in the application (abbreviated as KLP), which is described as

$$i_l = \underset{i \in B_l}{\operatorname{argmax}} \int_{-\infty}^{\infty} KL_i(\theta | \hat{\theta}) L(\hat{\theta}_l) d\theta \quad (5)$$

where $KL_i(\theta | \hat{\theta})$ is computed as

$$KL_i(\theta | \hat{\theta}) = P_i(\hat{\theta}) \ln \left[\frac{P_i(\hat{\theta})}{P_i(\theta)} \right] + [1 - P_i(\hat{\theta})] \ln \left[\frac{1 - P_i(\hat{\theta})}{1 - P_i(\theta)} \right] \quad (6)$$

FIWL and KLP yield greater accuracy than MFI for a short test length (Chen et al., 2000).

Van der Linden (1998) proposed several Bayesian approaches using the full posterior distribution. The maximum posterior weighted information strategy (abbreviated as MPWI) aims to maximize the information over the updated posterior distribution of the trait level and is described as

$$i_l = \underset{i \in B_l}{\operatorname{argmax}} \int_{-\infty}^{\infty} I_i(\hat{\theta}_l) G(\hat{\theta}_l) d\theta \quad (7)$$

where $G(\theta)$ is the updated posterior distribution obtained by Bayes' theorem.

The maximum expected information strategy (abbreviated as MEI) takes into account the information associated with the posterior predictive distribution, which evaluates the probabilities of correctly and incorrectly answering the target item and is described as

$$i_l = \underset{i \in B_l}{\operatorname{argmax}} P_i(\hat{\theta}_l) I_i(\hat{\theta}_{u_i=1}) + (1 - P_i(\hat{\theta}_l)) I_i(\hat{\theta}_{u_i=0}) \quad (8)$$

where $\hat{\theta}_{u_i=1}$ and $\hat{\theta}_{u_i=0}$ represent the updated trait levels if item i is selected and is answered correctly and incorrectly, respectively. These two estimators are obtained by the method of maximum a posteriori (MAP) or expected a posteriori (EAP). For short tests, Bayesian methods can decrease the mean squared error of the trait-level estimate (Meijer & Nering, 1999).

MDP framework and its corresponding elements in CAT

A CAT process has six principal components: (1) an item bank, (2) the start of the test, (3) an estimation method, (4) an item selection strategy, (5) constraint management, and (6) the end of the test (Weiss & Kingsbury, 1984; Frey, 2023). The present research exclusively applies reinforcement learning (RL) to augment the item selection strategy. The other five components can still be implemented using classical IRT models and common methods of CAT. Item selection can be

summarized as a cyclical interaction process between the testing system and the examinees (see Fig. 1a). The information-based strategy receives the estimated trait levels, calculates the information index of each item from the well-calibrated item bank based on the formula, and finally selects the item with the maximum information index. The examinee (or simulated response model) responds to that item. The estimation method updates the trait level, completing one round of the interactive cycle. Throughout multiple rounds of interaction, the objective of CAT is to maximize the cumulative item information. Such an interaction process and task objective are closely aligned with the paradigm of reinforcement learning.

In RL, the MDP abstracts the mathematical framework of the sequential decision problem. The basic idea of the MDP is simply to capture the most important aspects of a real problem in which a virtual agent interacts with its environment to achieve an ultimate goal. When adopting the MDP in CAT, the virtual agent is responsible for selecting items. The input received by the agent is referred to as the state, which is used for decision-making, and the output is referred to as the action. In the context of CAT, the state input to the agent is the trait level, and the action outputted by the agent is the item. The entity interacting with the agent is generally referred to as the environment in RL. The input that the environment receives is the action, and it ultimately outputs the state. The environment in CAT includes the examinee who responds to the item (or the IRT model generating simulated responses) and the estimation method, which converts responses into trait levels. The transition of states is called the environment model in RL. In the MDP, in addition to the state, the environment outputs immediate feedback to the agent, which is called the reward (see Fig. 1b). The reward defines not only how good the action is but also the goal of the RL problem. In CAT, it is the item information obtained after the examinee responds to the item, which can be measured by a simple item information index such as Fisher information. In the MDP, the Q-function serves as an objective function for each decision in the long run. In the context of CAT, the Q-function measures the weighted expected sum of the item information accumulated from the selection of the current item to the end of the test. Hence, selecting the item with the highest value of the Q-function represents the optimal choice from a long-term perspective.

After describing the item selection process in CAT using the MDP framework, the role of the newly proposed strategy in the procedure does not change compared to traditional IRT-based strategies. The only difference lies in how to acquire the item selection strategy based on reinforcement learning algorithms.

The setup described above exhibits three advantages. First, the representation of the state aligns the selection strategy under the MDP framework with the design logic of traditional IRT-based strategies, which select items based

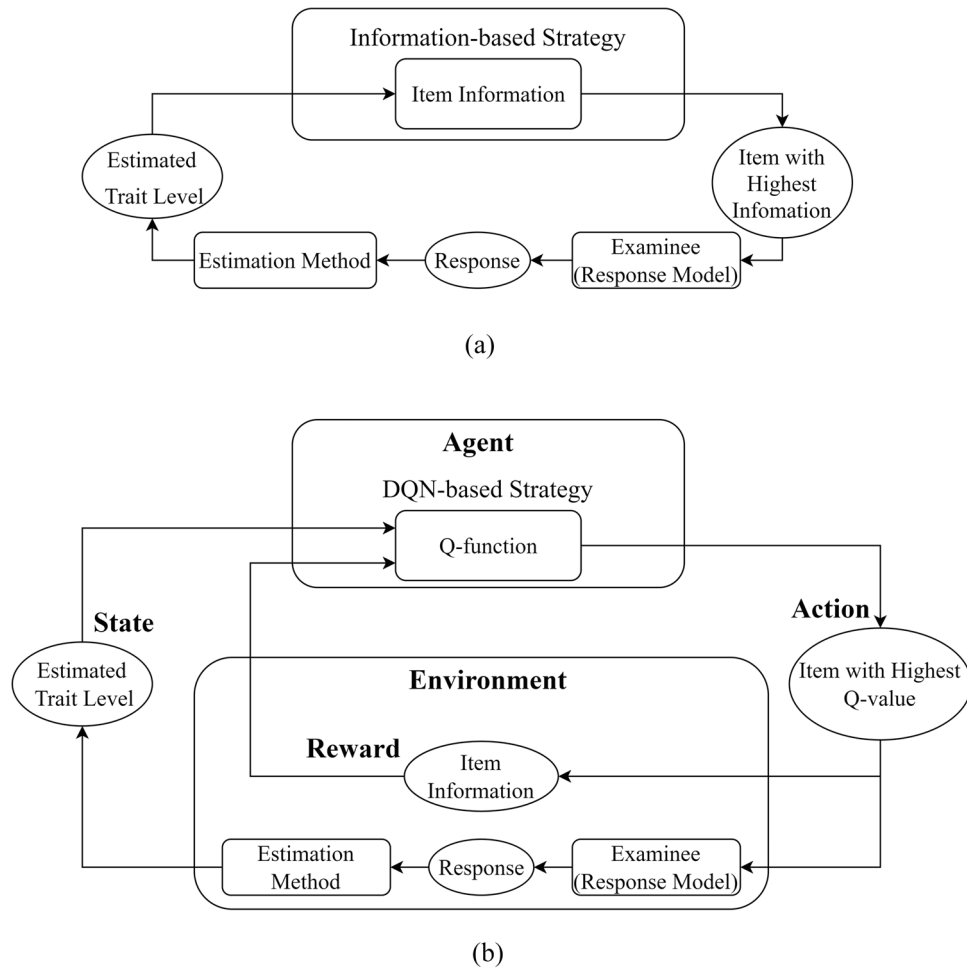


Fig. 1 Flowchart of the IRT-based item selection strategy and the RL-based item selection strategy in the MDP framework

on the estimated trait level in the sequence. Additionally, taking the most frequently used information measure as the evaluation method for each item in the MDP is in line with most information-based strategies, which take the Fisher information as the basis for improvement. Second, the key elements in the MDP can be calculated efficiently and accurately via IRT approaches such as MLE and IRT models for the environment and Fisher information for the reward, instead of adding complex modules to the MDP and increasing the computational complexity. Third, the objective of the agent in the MDP in making every decision is to pursue the maximization of cumulative rewards throughout the entire sequence, which is consistent with maximizing the test information.

The item selection strategy in CAT corresponds to the policy in the MDP. The policy, denoted as π , represents a set of rules for choosing actions in response to any kind of input state. That is, the policy π determines which item should be chosen for any given trait level in CAT. To utilize RL to maximize the sum of the information obtainable from the selection

of this item through to the end of the test, it is necessary to employ the Q-function. Suppose fixed-length CAT with L items is used. The estimated trait level when selecting item i_l is denoted as $\hat{\theta}_l$. The Fisher information brought by item i_l is $I_{i_l}(\hat{\theta}_l)$. After selecting the remaining $L - l$ items according to strategy π , the expectation of the weighted sum of the item information is measured by the Q-function. The Q-function theoretically considers all subsequent possibilities of item sequences based on policy π ; that is,

$$\begin{aligned} Q_{\pi}(\hat{\theta}_l, i_l) &= \mathbb{E}_{\pi} \left[I_{i_l}(\hat{\theta}_l) + \sum_{k=l+1}^L \gamma^{k-l} \cdot I_{i_k}(\hat{\theta}_k) | \hat{\theta}_l, i_l \right] \\ &= \mathbb{E}_{\pi} \left[I_{i_l}(\hat{\theta}_l) + \gamma \cdot \max_{i \in B_{l+1}} Q_{\pi}(\hat{\theta}_{l+1}, i) | \hat{\theta}_l, i_l \right] \end{aligned} \quad (9)$$

where γ is called the discount factor. Its value range is (0,1). The discount factor controls how much our agent discounts future rewards when making a decision. Larger γ values correspond to greater importance of future rewards. In CAT, the Q-function not only evaluates the efficacy of selecting

items for a given trait level, that is, the cumulative information obtained in subsequent testing for each choice, but also assesses the quality of the strategy π , as each subsequent test sequence will proceed in accordance with π . To obtain the ideal item selection strategy, it becomes essential to identify the optimal policy. The optimal policy, denoted as π^* , should always have the highest Q-value among all the policies corresponding to the pair of $\hat{\theta}_l$ and i_l . The Q-function of π^* is defined as

$$Q_{\pi^*}(\hat{\theta}_l, i_l) = \max_{\pi} Q_{\pi}(\hat{\theta}_l, i_l) \quad (10)$$

Once the optimal policy π^* is obtained, for any trait level, π^* can select the item with the highest Q-value, thus theoretically finding the optimal item selection strategy in CAT. The ideal strategy under the MDP is to always select the item with the largest Q-value of Q_{π^*} , described as

$$i_l = \underset{i \in B_l}{\operatorname{argmax}} Q_{\pi^*}(\hat{\theta}_l, i) \quad (11)$$

Next, there are two issues to address. First, the Q-values for all possible trait levels and items must be accurately determined. Second, the maximum Q-values must be identified for all trait levels and items, which represent the optimal item selection strategy.

Because $\hat{\theta}$ is a continuous variable that can take on an infinite number of values and because the size of B_l can also be large, the Q-function is likely a high-dimensional nonlinear function with infinite inputs and outputs. A simple linear model cannot fit this complex function with sufficient precision. In sequential tests such as CAT, it is impractical to collect all the sequences to find the optimal policy for all situations, as there is a substantial number of possibilities for the trajectory of the estimated trait levels. To solve such a problem, a well-designed training algorithm is needed, in addition to a neural network structure that can find the optimal policy π^* and fit its Q-function Q_{π^*} with a limited amount of data.

New item selection strategy based on the DQN algorithm

Similar to IRT-based strategies, a Q-function is employed for item selection in CAT, which takes continuous variables $\hat{\theta}$ as inputs and outputs the Q-values for each available item in the item bank. The relationship between the input and output in this scenario is a complex nonlinear function. DQN is especially suitable for discrete action spaces and complex Q-functions (Mnih et al., 2013, 2015). The new strategy based on DQN needs to fit the Q-function of the optimal strategy, which calculates the expectation of the sum of the information of all possible subsequent sequences given any

pair of $\hat{\theta}_l$ and i_l and finds the sequence with the most information under each combination. If every possibility is calculated to the end of the test according to an exhaustive method, such as MEI, the amount of calculation is too large, and the time and memory consumption are unacceptable. Therefore, DQN uses a neural network to explore and learn the Q-function for the optimal strategy from limited item selection records.

DQN training is divided into two parts: policy selection and policy evaluation, corresponding to exploring the optimal strategy and fitting the Q-function of this optimal strategy, respectively. During policy selection, DQN employs the ϵ -greedy method, selecting items randomly with a probability ϵ of exploring potentially better strategies, while selecting the item with the highest Q-value with a probability of $1 - \epsilon$ to make the optimal choice. DQN gradually builds its knowledge from scratch, accumulating experience to learn the most effective ways of selecting items. The ϵ -greedy method in DQN mirrors the balance between exploration and exploitation, akin to the process of acquiring new knowledge and consolidating existing knowledge. As long as there is adequate exploration of the state space and significant accumulation of data regarding item selections for different trait levels, the accuracy estimation of the Q-function for the optimal strategy will not be compromised, even if the data come from a random strategy (Zai & Brown, 2020, pp. 87). During policy evaluation, DQN adopts two techniques to make the Q-network better converge to the optimal Q-function: the target network and experience replay (Mnih et al., 2013, 2015). DQN designs two networks with the same structure: the action-value Q-network (denoted as Q_{action}) to fit the optimal Q-function and take the actions, and the target Q-network (denoted as Q_{target}) to compute the prediction target of Q_{action} . Utilizing a separate Q_{target} to generate predictive targets and periodically copying Q_{action} to the target network can make the Q-function converge more steadily and make the predictions more accurate, as well as reducing the complexity of training. From a psychometric perspective, the Q_{target} can be viewed as a stable, reliable validation test, while the Q_{action} resembles a new version of the test that is being calibrated against this benchmark. The relationship between two Q-networks in the CAT scenario is described as

$$Q_{action}(\hat{\theta}_l, i_l; \omega) \equiv I_{i_l}(\hat{\theta}_l) + \gamma \cdot \max_{i \in B_{l+1}} Q_{target}(\hat{\theta}_{l+1}, i; \omega') \quad (12)$$

where ω are the parameters of Q_{action} and ω' are the parameters of Q_{target} . The parameters of Q_{target} (i.e., ω') are copied from Q_{action} (i.e., ω) after Q_{action} has been updated several times. To efficiently and accurately train Q_{action} , another important ingredient of the DQN algorithm is experience replay. The item selection data accumulated in policy selection are integrated in the form $e_l = \{\hat{\theta}_l, i_l, I_{i_l}(\hat{\theta}_l), \hat{\theta}_{l+1}\}$ and

stored in a memory bank $D = \{e_1, \dots, e_{N_D}\}$, which can hold N_D records. The data in the replay memory with limited capacity are continuously refreshed, thereby stabilizing the training process of DQN by disrupting temporal correlations and leveraging past experiences. Then, a mini-batch of m records $\{\hat{\theta}_l, i_l, I_{i_l}(\hat{\theta}_l), \hat{\theta}_{l+1}\}$ are sampled from memory bank D and used to update Q_{action} . Employing mini-batches not only reduces the dependency among records but also accelerates the updating process of Q_{action} , aligning with standard practices for optimizing neural networks. Experience replay acts as a teacher in planning the learning material by updating and randomly drawing the material learned by Q_{action} . In summary, the algorithm for training DQN is as follows:

Initialize the action-value network Q_{action} with random weights ω , the target network Q_{target} with random weights $\omega' = \omega$, and the replay memory D with size N_D . For the $N_{training}$ examinees, repeat the following cycle (all elements within a single cycle belong to the same examinee $n = 1, 2, \dots, N$):

(0) Generate the initial trait level $\hat{\theta}_1$.

For test sequence $l = 1, 2, \dots, L$:

- (1) Input $\hat{\theta}_l$ into Q_{action} to obtain the Q-values of all items, and retain only those corresponding to B_l .
- (2) Select item i_l randomly from B_l with probability ϵ ; otherwise, select item $i_l = \underset{i \in B_l}{\operatorname{argmax}} Q_{action}(\hat{\theta}_l, i; \omega)$.
- (3) Calculate the Fisher information $I_{i_l}(\hat{\theta}_l)$ and obtain the updated trait level $\hat{\theta}_{l+1}$ by the estimation method.
- (4) Store $\{\hat{\theta}_l, i_l, I_{i_l}(\hat{\theta}_l), \hat{\theta}_{l+1}\}$ in D .

When the number of records stored in D is greater than the batch size m , do the following:

- (5) Sample a mini-batch of m records $\{\hat{\theta}_l, i_l, I_{i_l}(\hat{\theta}_l), \hat{\theta}_{l+1}\}$ from D .
- (6) Update parameters ω of Q_{action} by optimizing the loss function

$$\begin{cases} [I_{i_l}(\hat{\theta}_l) - Q_{action}(\hat{\theta}_l, i_l; \omega)]^2 & \text{if } l = L \\ [I_{i_l}(\hat{\theta}_l) + \gamma \cdot \underset{i \in B_{l+1}}{\operatorname{max}} Q_{target}(\hat{\theta}_{l+1}, i; \omega') - Q_{action}(\hat{\theta}_l, i_l; \omega)]^2 & \text{otherwise} \end{cases} \quad (13)$$

- (7) For every τ times Q_{action} updates, copy weights $\omega' = \omega$.

The MDP proposed in this study combines the concepts of IRT that allow the DQN to learn from real data or simulated data. The former type of data means that $\hat{\theta}_{l+1}$ in step (3) is estimated from a human-generated response or a past response record generated by real people. The latter refers to responses generated through IRT models or other

simulation methods. Both types of data or examinees are available, with their own fidelity and cost advantages, and they can be used separately or in combination for training DQN. As seen in step (4), the records for training DQN only contain the i_l labeling the item numbers in the bank and not the item parameters, so each Q-network only corresponds to the item bank from which the records originate. If the item bank changes, the optimal choice for the same trait level may change. Therefore, it is recommended to modify the optimal Q_{action} that was initially developed for the original item bank. Such adjustments entail integrating new data corresponding to the newly added items in combination with the response data from the items that are retained. Then Q_{action} is fine-tuned or retrained using the same training algorithm, but now applied to the newly incorporated data. In step (6), there are two versions of the loss function for updating Q_{action} : the first is used when the CAT reaches the fixed-length L , and the second occurs while the testing is still in progress. Inside the loop from steps (1) to (7), all elements belong to examinee n . Only the elements in the mini-batch during steps (5) to (6) are randomly drawn from memory, and they may come from any examinee at any length. Moreover, this study assumes that the CAT is fixed-length. For variable-length CAT, we only need to adjust the termination rule for the loop, along with the termination criteria in the first version of the loss function in step (6).

According to Fig. 2, it can be observed that, compared with using DQN for item selection after training (as shown in Fig. 1b), the role of the environment implemented by IRT methods remains unchanged during the training of the DQN item selection strategy. This implies that the training of the DQN-based item selection strategy can be conducted by either an IRT model or real responses, with the training process being entirely consistent.

Simulation study

Designs

The new DQN-based item selection strategy was compared with five traditional IRT-based strategies (MFI, FIWL, KLP, MPWI, and MEI) regarding selection accuracy through a simulation study. 3PLM was used to simulate the item parameters and calculate the Fisher information. Considering that parameters a and b of the items are usually correlated (Wingersky & Lord, 1984; Chang et al., 2001), two types of item banks were generated, one with uncorrelated a and b parameters ($r_{ab} = 0$) and the other with correlated parameters ($r_{ab} = 0.5$). The distribution of item parameters, the number of item banks, and the number of items refer to the existing research (Barrada, Abad, & Veldkamp, 2009a, Barrada, Olea, et al., 2009b, 2010; Sorrel et al., 2020). Each

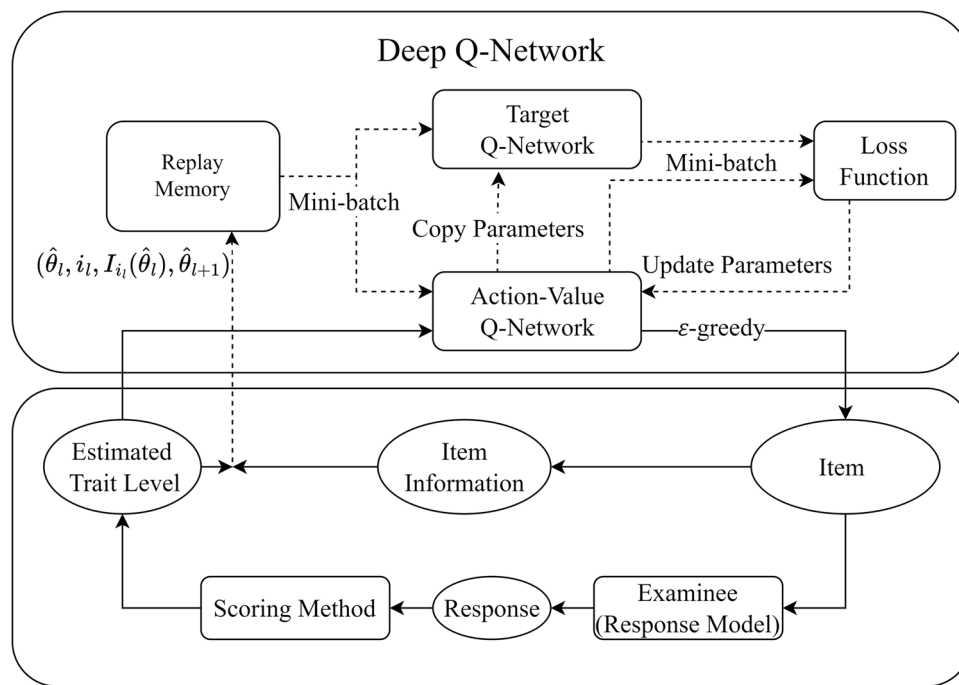


Fig. 2 Flowchart of the training process for the DQN-based item selection strategy

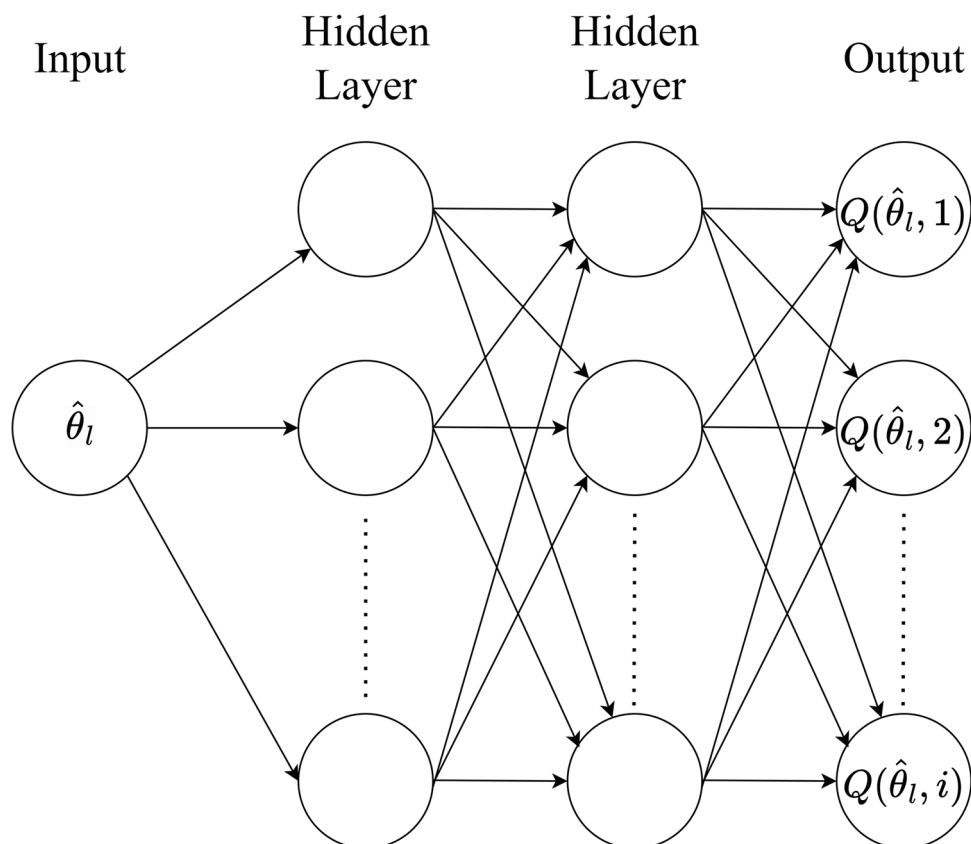


Fig. 3 The structure of the Q-network

type consisted of 10 item banks, with each bank comprising 500 items. The parameter distributions were $a \sim N(1.2, 0.25)$, $b \sim N(0, 1)$ and $c \sim N(0.25, 0.02)$. For each item bank, the trait levels of 5000 examinees were randomly generated from the distribution $N(0, 1)$. The initial trait level was randomly chosen from a uniform interval $[-0.5, 0.5]$. Dodd's (1990) method was used until correct and incorrect responses were produced for each examinee. If all responses were correct, $\hat{\theta}_{l+1}$ was increased by $(b_{max} - \hat{\theta}_l)/2$; alternatively, if all responses were incorrect, $\hat{\theta}_{l+1}$ was decreased by $(\hat{\theta}_l - b_{min})/2$, where b_{max} and b_{min} represent the maximum and minimum values of the difficulty parameter within the item bank, respectively. Then, MLE was applied to estimate trait levels within the range $[-4, 4]$. Four different test lengths, 10, 20, 30, and 40 items, were used. The prior involved in the two Bayesian strategies was the standard normal distribution, and MAP was used to estimate the predictive trait levels for MEI. The selection of items for the five IRT-based strategies, generation of responses, and estimation of abilities were implemented by the functions provided in the R package catR (Magis & Barrada, 2017).

Evaluation criteria

Four commonly used metrics in CAT were used to evaluate the recovery of the examinees' true trait levels (He et al., 2014). The evaluation metrics are defined as follows:

(1) BIAS

$$BIAS = \frac{1}{N} \sum_{n=1}^N (\hat{\theta}_n - \theta_n) \quad (14)$$

where N is the number of examinees, and $\hat{\theta}_n$ and θ_n are the estimated trait level and true trait level of the n th examinee, respectively.

(2) Mean absolute error (MAE)

$$MAE = \frac{1}{N} \sum_{n=1}^N |\hat{\theta}_n - \theta_n| \quad (15)$$

(3) Root mean square error (RMSE)

$$RMSE = \sqrt{\frac{1}{N} \sum_{n=1}^N (\hat{\theta}_n - \theta_n)^2} \quad (16)$$

The smaller the values of the three indicators above, the more accurate the strategy is.

(4) The correlation between the true trait level and the final estimated value is

$$r_{\theta, \hat{\theta}} = \frac{\sum_{n=1}^N (\theta_n - \bar{\theta})(\hat{\theta}_n - \bar{\hat{\theta}})}{S_{\theta} S_{\hat{\theta}}} \quad (17)$$

where $\bar{\theta}$ and S_{θ} are the mean and standard deviation of the true trait levels of all n examinees, and $\bar{\hat{\theta}}$ and $S_{\hat{\theta}}$ are the mean and variance of the estimated trait levels of all n examinees. A larger value of this indicator corresponds to a better strategy.

Implementation of the DQN algorithm

The same network structure and training process were used to train the optimal Q-network for each of the 20 simulated item banks. DQN puts no requirements on the structure of Q-networks. In this study, a multilayer fully connected network with a nonlinear activation function is used. There are two hidden layers between the input and output, and each hidden layer uses a nonlinear activation function called the rectified linear unit (ReLU), where $ReLU(x) = \max(0, x)$. The structure of the Q-network is presented in Fig. 3 (same as in Tan et al., 2019). The dimension of the Q-network's input equals 1 (for $\hat{\theta}$), and that of the output equals 500 (the size of the item bank). The two hidden layers have 50 and 30 neurons. This Q-network has $2 \times 50 + 51 \times 30 + 31 \times 500 = 17,130$ parameters in total (each layer adds a neuron before connecting with the next layer, which can be seen as the intercept in regression). To ensure that the information calculated by the Q-network is positive, all network parameters are constrained to be positive.

In the simulation study, the training of DQN using simulated response data will be demonstrated. Consistent with the IRT model for calibrating item parameters, the 3PLM is used to generate response data for training DQN. Based on deep learning training experience, the quantity of data contributes to the performance of the neural networks: the larger the scale of data in training the DQN, the greater the likelihood of performance improvement (Goodfellow et al., 2016, pp. 19–20). Both human examinees and real response records are difficult and expensive to obtain in real-world applications of CAT, especially if the size is as large as the one used to compare the strategies in the above simulation (5000 examinees, denoted as $N_{testing}$). Furthermore, using real response records requires a large quantity of data to ensure that responses or updated traits for all available items selected by DQN can be found in the dataset.

Another factor that may impact the performance of DQN is the distribution of the true trait levels of examinees in the training phase. This determines the distribution of $\hat{\theta}$ in the training data and further influences whether the optimal strategy learned by the DQN during training can be generalized to the testing phase. To enhance the generalizability of deep learning models, it is recommended that the training

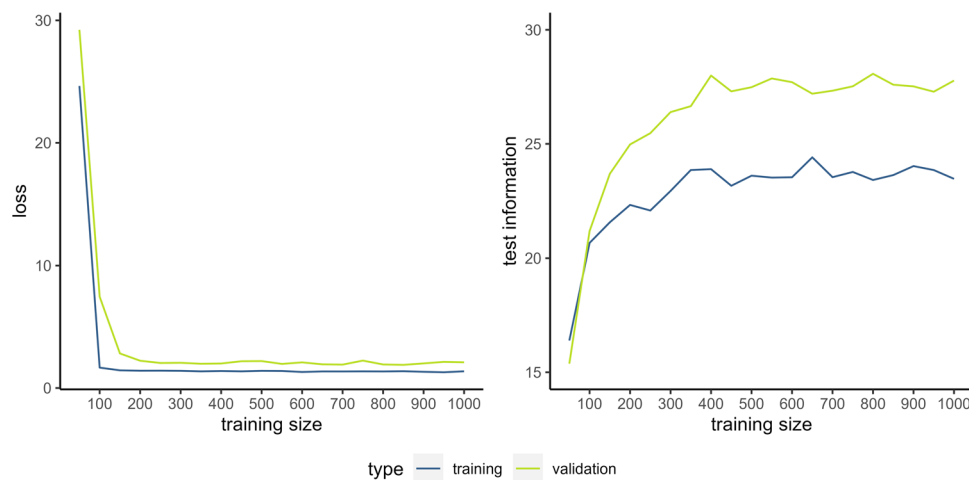


Fig. 4 Changes in the loss and test information during the training and validation phases with increasing sample size

and testing data have identical distributions (Goodfellow et al., 2016, pp. 109–116). To simulate scenarios where the abilities of target examinees are known, the distribution of trait levels for generating training data was set to match the standard normal distribution used in the testing phase. However, in practical scenarios, determining the exact distribution of the target examinees' trait levels might be challenging. To ensure that DQN encounters training data across the full spectrum of trait levels, a uniform distribution ranging from -3 to 3 was specified for generating examinees for DQN training. In summary, using the same standard normal distribution as in the testing phase can be considered the ideal training situation, while a uniform distribution serves as a remedial approach for cases where little is known about the trait levels of the testing examinees.

To avoid overfitting and to obtain a DQN model with good generalizability, a validation phase was designed during the training process. For the purpose of differentiation, the phase in which the Q-network is updated following the algorithm described above is called the training phase. Every time $N_{interval}$ examinees' records (where $N_{interval}$ is a divisor of $N_{training}$) are used to update the network Q_{action} in the training phase, $N_{validation}$ examinees with the same distribution of trait levels as in the test phase interact with the current optimal Q_{action} to complete the L -item CAT. The loss and reward of DQN and the RMSE of selection were recorded throughout the training and validation phases to monitor the convergence of the Q-network and explore the effect of the quantity of data. In addition, the discount factor γ also has a positive effect on model convergence and generalization performance (Schulman et al., 2015; Amit et al., 2020). Improper values of γ make it difficult for the Q-network to converge, and the policy learned is not globally optimal. Different item banks and groups of examinees may lead to different γ values in the optimal Q-function. For each bank and type of distribution, four values of γ (0.1, 0.3, 0.6,

and 0.9) were taken to obtain the optimal Q_{action} . To quickly find the best value and prevent overfitting, the early-stopping technique was used in the training phase. If the loss was high and did not decrease or was very different for the training and validation, training was halted to adjust the value of γ . After the best value of γ was determined, the training continued until 1000 examinees had completed the 40-item test.

The replay memory size was 1000 (N_D). Every 40 times (τ) that Q_{action} was updated, Q_{target} was updated once, which means that Q_{target} was updated once after each examinee completed the 40-item test. The mini-batch size (m) was set as 128. The optimization algorithm used for training the network was Adam (Kingma & Ba, 2014), and the learning rate was 0.001. The trait distribution of examinees for the validation phases was the standard normal distribution. Each examinee finished the 40-item (L) test in both phases. $N_{interval}$ was 50, and $N_{validation}$ was 200. A new optimal Q_{action} was saved if it had lower BIAS and RMSE scores on average and higher rewards during the 40-item tests than the previous optimal network. The item selection strategy based on DQN takes the action with the maximal Q-value calculated by the optimal Q_{action} , as described in Eq. (11). Then, 5000 examinees were tested by DQN in the testing stage and compared with the traditional strategies. The ϵ -greedy value was set to 0.1 in the training phase and 0 in the validation and testing phases. All training and testing related to DQN were implemented using the Python deep learning programming framework PyTorch (Paszke et al., 2019), run on a GPU and accelerated by CUDA and cuDNN.² The CPU was an Intel Core i7-7700 @3.60 GHz, and the GPU was an Nvidia GeForce GTX 1080Ti with memory size of 8192 MB.

² The Python code for training and testing DQN and the R code of CAT based on the traditional strategies for both studies are shared on the OSF platform.

Table 1 Recovery of trait levels for different item selection strategies under the banks with uncorrelated a and b parameters

Test length	Method	BIAS	RMSE	MAE	$r_{\theta,\hat{\theta}}$
10	DQN (normal)	−0.038	0.415	0.313	0.950
	DQN (uniform)	−0.014	0.348	0.276	0.964
	MFI	0.037	0.493	0.371	0.899
	KLP	0.018	0.443	0.343	0.914
	FIWL	0.021	0.459	0.354	0.909
	MPWI	0.020	0.456	0.347	0.912
	MEI	0.025	0.455	0.347	0.911
20	DQN (normal)	−0.074	0.278	0.208	0.977
	DQN (uniform)	−0.061	0.254	0.204	0.981
	MFI	0.012	0.331	0.257	0.950
	KLP	0.012	0.322	0.250	0.952
	FIWL	0.011	0.324	0.253	0.952
	MPWI	0.011	0.320	0.250	0.953
	MEI	0.012	0.324	0.253	0.952
30	DQN (normal)	−0.034	0.244	0.187	0.982
	DQN (uniform)	−0.001	0.207	0.167	0.987
	MFI	0.007	0.277	0.216	0.965
	KLP	0.007	0.272	0.212	0.965
	FIWL	0.008	0.274	0.214	0.965
	MPWI	0.006	0.272	0.213	0.966
	MEI	0.007	0.273	0.213	0.965
40	DQN (normal)	−0.008	0.227	0.176	0.986
	DQN (uniform)	−0.004	0.207	0.169	0.989
	MFI	0.005	0.245	0.191	0.972
	KLP	0.006	0.243	0.190	0.972
	FIWL	0.006	0.245	0.191	0.972
	MPWI	0.006	0.243	0.190	0.972
	MEI	0.005	0.244	0.191	0.972

Note. DQN (normal): Deep Q-network trained with a standard normal distribution. DQN (uniform): Deep Q-network trained with a uniform distribution. MFI: Maximize the Fisher information. KLP: Maximize the Kullback–Leibler information weighted by the likelihood. FIWL: Maximize the Fisher information weighted by the likelihood. MPWI: Maximize the posterior weighted information. MEI: Maximize the expected information

Results

The effects of the correlation of parameters a and b and the distribution of trait levels for training DQN are so small that the lines separated by $r_{ab} = 0/0.5$ and the type of distribution almost overlap (these two factors are not shown separately in Fig. 4). For the optimal Q-networks of all item banks, the loss of the training phase reduces quickly and is maintained at a very low level when training on only 100 examinees. After the Q-network interacts with another 100 examinees, the loss of the validation phase also drops to a very low level and does not increase. Since the two phases are very close, there is no overfitting.

Table 2 Recovery of trait levels for different item selection strategies under banks with correlated a and b parameters

Test length	Method	BIAS	RMSE	MAE	$r_{\theta,\hat{\theta}}$
10	DQN (normal)	0.003	0.450	0.375	0.961
	DQN (uniform)	0.015	0.438	0.347	0.947
	MFI	0.018	0.554	0.404	0.878
	KLP	−0.003	0.493	0.371	0.898
	FIWL	0.011	0.541	0.398	0.881
	MPWI	−0.006	0.507	0.374	0.893
	MEI	0.003	0.512	0.377	0.890
20	DQN (normal)	−0.013	0.299	0.239	0.978
	DQN (uniform)	0.012	0.301	0.242	0.973
	MFI	0.002	0.368	0.277	0.940
	KLP	−0.001	0.349	0.268	0.945
	FIWL	0.004	0.361	0.275	0.942
	MPWI	−0.003	0.352	0.267	0.944
	MEI	0.001	0.356	0.271	0.942
30	DQN (normal)	−0.011	0.238	0.180	0.984
	DQN (uniform)	−0.005	0.237	0.175	0.977
	MFI	−0.001	0.301	0.229	0.959
	KLP	−0.001	0.291	0.225	0.961
	FIWL	0.000	0.299	0.229	0.959
	MPWI	−0.003	0.296	0.225	0.959
	MEI	−0.001	0.296	0.227	0.959
40	DQN (normal)	−0.037	0.250	0.190	0.986
	DQN (uniform)	−0.004	0.224	0.168	0.982
	MFI	−0.002	0.266	0.203	0.967
	KLP	−0.002	0.259	0.200	0.968
	FIWL	−0.001	0.263	0.203	0.968
	MPWI	−0.004	0.264	0.201	0.967
	MEI	−0.001	0.263	0.201	0.967

Note. DQN (normal): Deep Q-network trained with a standard normal distribution. DQN (uniform): Deep Q-network trained with a uniform distribution. MFI: Maximize the Fisher information. KLP: Maximize the Kullback–Leibler information weighted by the likelihood. FIWL: Maximize the Fisher information weighted by the likelihood. MPWI: Maximize the posterior weighted information. MEI: Maximize the expected information

Under these two phases, the unweighted cumulative reward (test information of the 40-item test) rises rapidly when training on the first 200 examinees and then grows gradually as the amount of training data increases. This is evidence that DQN will continue to search for a better policy with increasing amounts of training data. The low loss throughout the training phase indicates that whenever a new optimal policy is found, the Q-network converges to the Q-function of that policy quickly and accurately. The test information of the validation phase is always higher than that of the training phase because a 10% probability of random selection ($\epsilon = 0.1$) is set during the training phase, while items with the largest Q-values calculated by the optimal

Q_{action} are definitively selected ($\varepsilon = 0$) during validation. When the values of γ are set to 0.1, 0.3, 0.6, and 0.9, the proportions of achieving the best model are 40%, 45%, 15%, and 0%, respectively.

In the simulation study, the results are presented by type of item bank ($r_{ab} = 0 / 0.5$), and all evaluation metrics are calculated as the average values across the 10 item banks for each type. Compared to the five classical IRT-based selection strategies, the DQN-based strategy (hereinafter abbreviated as DQN) in the testing phase shows lower RMSE and MAE values and a higher $r_{\theta, \hat{\theta}}$ at all four test lengths under the two kinds of item banks (shown in Tables 1 and 2). More importantly, the DQN trained with a uniform distribution performs better than the model with a standard normal

distribution. In terms of RMSE and MAE, DQN with both types of distributions can reach the measurement accuracy using only 20 items, whereas traditional strategies need 30 items. The traditional strategies tend to select the same items when the test length increases, which makes the test accuracy of a long test very similar (Meijer & Nering, 1999). When CAT increases the number of items from 30 to 40, the DQN continues to reduce the RMSE, while the RMSE under the traditional strategies hardly changes. In terms of BIAS, DQN has slightly larger values within the acceptable range for short test lengths, and maintains the same level of BIAS as the traditional strategy in a longer 40-item test.

In the testing phase, examinees are categorized into five groups based on their trait levels. This classification

Table 3 RMSE values of trait levels for different item selection strategies and examinees with different true trait levels under banks with uncorrelated a and b parameters

Test length	Method	Range of true trait levels				
		(−3, −1)	(−1, −0.5)	(−0.5, 0.5)	(0.5, 1)	(1, 3)
10	DQN (normal)	0.527	0.400	0.304	0.303	0.468
	DQN (uniform)	0.399	0.316	0.323	0.302	0.342
	MFI	0.574	0.493	0.469	0.467	0.484
	KLP	0.528	0.423	0.408	0.416	0.471
	FIWL	0.520	0.445	0.438	0.436	0.475
	MPWI	0.568	0.430	0.397	0.416	0.515
	MEI	0.572	0.445	0.407	0.412	0.480
20	DQN (normal)	0.406	0.231	0.207	0.200	0.314
	DQN (uniform)	0.290	0.247	0.244	0.206	0.252
	MFI	0.381	0.321	0.308	0.315	0.358
	KLP	0.378	0.307	0.297	0.305	0.346
	FIWL	0.375	0.314	0.302	0.304	0.348
	MPWI	0.380	0.307	0.295	0.298	0.348
	MEI	0.392	0.311	0.295	0.301	0.350
30	DQN (normal)	0.349	0.205	0.188	0.198	0.267
	DQN (uniform)	0.217	0.211	0.202	0.164	0.215
	MFI	0.321	0.267	0.255	0.261	0.302
	KLP	0.323	0.260	0.250	0.254	0.293
	FIWL	0.318	0.265	0.253	0.258	0.295
	MPWI	0.322	0.258	0.250	0.253	0.297
	MEI	0.328	0.262	0.250	0.250	0.296
40	DQN (normal)	0.293	0.203	0.183	0.197	0.252
	DQN (uniform)	0.222	0.209	0.198	0.168	0.219
	MFI	0.290	0.236	0.223	0.229	0.270
	KLP	0.290	0.233	0.223	0.226	0.261
	FIWL	0.287	0.235	0.227	0.229	0.264
	MPWI	0.289	0.232	0.222	0.225	0.265
	MEI	0.296	0.235	0.223	0.222	0.261

Note. DQN (normal): Deep Q-network trained with a standard normal distribution. DQN (uniform): Deep Q-network trained with a uniform distribution. MFI: Maximize the Fisher information. KLP: Maximize the Kullback–Leibler information weighted by the likelihood. FIWL: Maximize the Fisher information weighted by the likelihood. MPWI: Maximize the posterior weighted information. MEI: Maximize the expected information

aims to assess the test accuracy of all strategies for examinees within each group. The results in Tables 3 and 4 indicate that regardless of which distributions are used in the training phase, DQN has a lower RMSE than the other five IRT-based strategies for most groups under all test lengths and bank types. Training with the normal distribution decreases the RMSE of DQN, especially for examinees whose trait levels are close to the average. DQN trained with a uniform distribution has the lowest RMSE, especially for examinees with very high or low trait levels, compared to all other strategies including DQN trained with the normal distribution. For the BIAS for each group (see Tables 5 and 6), DQNs trained under both distributions are generally higher overall than traditional strategies. This is because DQN may show a certain

direction of BIAS under several item banks. For those banks where DQN produces a larger BIAS, the proportion of examinees with an absolute BIAS value between 0.1 and 0.3 averaged 30% to 35% for DQN, which is lower than that for MFI. However, the larger BIAS values for MFI have an equal probability of being either positive or negative, whereas for DQN, they are predominantly on one side of zero. This leads to a directional BIAS among the total 5000 examinees in that bank, and synthesizing the results from 10 item banks cannot completely offset this BIAS to zero.

To demonstrate the differences in performance for a particular examinee using MFI and DQN trained with a standard normal distribution, the item selections of a simulated examinee in the 40-item test under an item bank with $r_{ab} = 0$

Table 4 RMSE values of trait levels for different item selection strategies and examinees with different true trait levels under banks with correlated a and b parameters

Test length	Method	Range of true trait levels				
		(−3, −1)	(−1, −0.5)	(−0.5, 0.5)	(0.5, 1)	(1, 3)
10	DQN (normal)	0.624	0.429	0.384	0.355	0.412
	DQN (uniform)	0.578	0.372	0.424	0.371	0.350
	MFI	0.777	0.606	0.495	0.446	0.447
	KLP	0.690	0.512	0.433	0.406	0.434
	FIWL	0.744	0.574	0.494	0.448	0.442
	MPWI	0.754	0.543	0.425	0.407	0.423
	MEI	0.752	0.547	0.440	0.405	0.424
20	DQN (normal)	0.443	0.261	0.248	0.244	0.280
	DQN (uniform)	0.398	0.303	0.254	0.268	0.285
	MFI	0.520	0.379	0.327	0.300	0.321
	KLP	0.482	0.353	0.310	0.293	0.318
	FIWL	0.499	0.373	0.325	0.298	0.322
	MPWI	0.503	0.356	0.307	0.293	0.316
	MEI	0.506	0.372	0.310	0.291	0.316
30	DQN (normal)	0.370	0.187	0.179	0.181	0.256
	DQN (uniform)	0.362	0.213	0.194	0.185	0.199
	MFI	0.425	0.304	0.267	0.249	0.270
	KLP	0.400	0.294	0.261	0.243	0.267
	FIWL	0.416	0.305	0.266	0.246	0.272
	MPWI	0.424	0.296	0.259	0.244	0.268
	MEI	0.417	0.308	0.259	0.247	0.265
40	DQN (normal)	0.417	0.210	0.185	0.175	0.211
	DQN (uniform)	0.346	0.213	0.176	0.181	0.180
	MFI	0.376	0.267	0.234	0.218	0.241
	KLP	0.357	0.264	0.232	0.214	0.236
	FIWL	0.365	0.269	0.234	0.217	0.244
	MPWI	0.378	0.263	0.231	0.217	0.239
	MEI	0.369	0.272	0.229	0.218	0.239

Note. DQN (normal): Deep Q-network trained with a standard normal distribution. DQN (uniform): Deep Q-network trained with a uniform distribution. MFI: Maximize the Fisher information. KLP: Maximize the Kullback–Leibler information weighted by the likelihood. FIWL: Maximize the Fisher information weighted by the likelihood. MPWI: Maximize the posterior weighted information. MEI: Maximize the expected information

Table 5 BIAS values of trait levels for different item selection strategies and examinees with different true trait levels under banks with uncorrelated a and b parameters

Test length	Method	Range of true trait levels				
		(−3, −1)	(−1, −0.5)	(−0.5, 0.5)	(0.5, 1)	(1, 3)
10	DQN (normal)	−0.145	−0.127	0.031	−0.023	−0.028
	DQN (uniform)	−0.003	0.010	−0.051	0.001	0.032
	MFI	0.018	0.035	0.034	0.052	0.055
	KLP	0.016	0.016	0.015	0.016	0.029
	FIWL	0.016	0.017	0.021	0.020	0.028
	MPWI	0.001	0.018	0.014	0.022	0.054
	MEI	0.012	0.032	0.023	0.018	0.043
20	DQN (normal)	−0.179	−0.094	−0.055	−0.045	−0.023
	DQN (uniform)	−0.066	−0.078	−0.067	−0.053	−0.034
	MFI	−0.006	0.006	0.010	0.022	0.034
	KLP	0.002	0.007	0.011	0.015	0.025
	FIWL	0.004	0.006	0.011	0.013	0.024
	MPWI	−0.004	0.017	0.008	0.019	0.023
	MEI	−0.004	0.016	0.012	0.013	0.025
30	DQN (normal)	−0.133	−0.047	−0.008	−0.013	−0.008
	DQN (uniform)	0.008	−0.006	−0.008	0.008	0.005
	MFI	−0.009	0.005	0.005	0.015	0.026
	KLP	−0.003	0.001	0.007	0.011	0.018
	FIWL	0.000	0.002	0.008	0.012	0.021
	MPWI	−0.011	0.009	0.005	0.013	0.018
	MEI	−0.008	0.007	0.007	0.009	0.020
40	DQN (normal)	−0.088	−0.030	0.009	0.019	0.023
	DQN (uniform)	−0.041	−0.044	−0.044	−0.033	−0.032
	MFI	−0.010	0.002	0.003	0.011	0.021
	KLP	−0.005	−0.001	0.006	0.010	0.018
	FIWL	−0.003	0.001	0.007	0.011	0.016
	MPWI	−0.008	0.008	0.005	0.012	0.014
	MEI	−0.009	0.005	0.004	0.007	0.016

Note. DQN (normal): Deep Q-network trained with a standard normal distribution. DQN (uniform): Deep Q-network trained with a uniform distribution. MFI: Maximize the Fisher information. KLP: Maximize the Kullback–Leibler information weighted by the likelihood. FIWL: Maximize the Fisher information weighted by the likelihood. MPWI: Maximize the posterior weighted information. MEI: Maximize the expected information

were analyzed. The true trait level of this simulated examinee is -1.361 . As depicted in Fig. 5, DQN rapidly reduces RMSE within the first five items, and consistently presents a lower RMSE compared to MFI from the 10th item onwards (see Fig. 5a). According to the calculation of Fisher item information based on the current estimated trait levels, MFI mostly selects items with higher Fisher information (see Fig. 5b). However, when calculating Fisher information based on the true trait level, the items selected by DQN had significantly higher Fisher information (see Fig. 5c). When all items selected by both strategies were input into the optimal Q-network of the item bank (see Fig. 5d), it was found that the items chosen by DQN had higher Q-values than those selected by MFI. Only 5 out of the 40 items selected

by the two strategies overlapped. These results indicate that the Q-values calculated by the optimal Q-network can identify items that are suitable to the true trait level; thus the DQN-based strategy can plan a globally optimal sequence of the entire test.

Empirical study based on the real bank and responses

The simulation study assessed the feasibility and effectiveness of the proposed DQN-based item selection strategy. In this section, we demonstrate how to train a DQN-based strategy using response data generated by real examinees

Table 6 BIAS values of trait levels for different item selection strategies and examinees with different true trait levels under banks with correlated a and b parameters

Test length	Method	Range of true trait levels				
		(−3, −1)	(−1, −0.5)	(−0.5, 0.5)	(0.5, 1)	(1, 3)
10	DQN (normal)	−0.088	−0.025	−0.040	−0.033	0.054
	DQN (uniform)	−0.003	0.010	−0.051	0.001	0.032
	MFI	−0.024	0.001	0.029	0.032	0.035
	KLP	−0.025	−0.010	0.003	0.001	0.005
	FIWL	−0.018	0.002	0.018	0.024	0.020
	MPWI	−0.040	−0.014	0.004	0.007	−0.003
	MEI	−0.015	0.012	0.007	0.001	0.005
20	DQN (normal)	−0.052	0.011	−0.015	−0.029	0.028
	DQN (uniform)	−0.066	−0.078	−0.067	−0.053	−0.034
	MFI	−0.040	−0.002	0.009	0.013	0.020
	KLP	−0.019	−0.007	0.004	0.005	0.008
	FIWL	−0.032	0.002	0.011	0.010	0.017
	MPWI	−0.023	−0.002	−0.001	0.000	0.007
	MEI	−0.012	0.010	0.001	0.000	0.006
30	DQN (normal)	−0.043	0.012	−0.008	−0.018	0.060
	DQN (uniform)	0.008	−0.006	−0.008	0.008	0.005
	MFI	−0.029	−0.006	0.004	0.006	0.014
	KLP	−0.016	−0.006	0.002	0.001	0.007
	FIWL	−0.030	−0.003	0.005	0.005	0.013
	MPWI	−0.022	−0.001	−0.001	−0.002	0.008
	MEI	−0.014	0.004	0.000	−0.001	0.007
40	DQN (normal)	−0.057	−0.001	0.006	0.003	0.065
	DQN (uniform)	−0.041	−0.044	−0.044	−0.033	−0.032
	MFI	−0.026	−0.006	0.003	0.004	0.012
	KLP	−0.015	−0.007	0.001	0.000	0.008
	FIWL	−0.022	−0.004	0.002	0.004	0.010
	MPWI	−0.022	−0.002	−0.002	−0.002	0.007
	MEI	−0.014	0.003	−0.001	0.002	0.006

Note. DQN (normal): Deep Q-network trained with a standard normal distribution. DQN (uniform): Deep Q-network trained with a uniform distribution. MFI: Maximize the Fisher information. KLP: Maximize the Kullback–Leibler information weighted by the likelihood. FIWL: Maximize the Fisher information weighted by the likelihood. MPWI: Maximize the posterior weighted information. MEI: Maximize the expected information

while also comparing the accuracy of traditional strategies and the new strategy.

Designs

The data are from a large-scale medical test in China. The item bank consists of 371 multiple-choice items, each with four options, calibrated using the 3PLM. The descriptive statistics of item parameters are shown in Table 7. Training the DQN with real response data also requires a large amount of $e_i = \{\hat{\theta}_i, i_i, I_i(\hat{\theta}_i), \hat{\theta}_{i+1}\}$, meaning that for any item i_i selected by the current optimal DQN-based strategy at any trait level $\hat{\theta}_i$, there must be a corresponding response for estimating $\hat{\theta}_{i+1}$. Therefore, we used examinees who had

responses for all items (excluding those with all correct or incorrect answers). To maintain consistency with the number of examinees in the training and testing phases of the simulation study, we utilized response data from 5000 examinees for training the DQN. Another 5000 examinees were used to test the accuracy of DQN and compare its performance with traditional strategies. The trait levels estimated from responses to all items were treated as the true values. The four metrics described in Section "Evaluation criteria" (BIAS, RMSE MAE, and r_{ab}) were used to evaluate the trait levels of each strategy across four lengths of CAT (10, 20, 30, and 40). During both the training and testing phases, real response data were used rather than responses generated by the 3PLM model. The initial trait

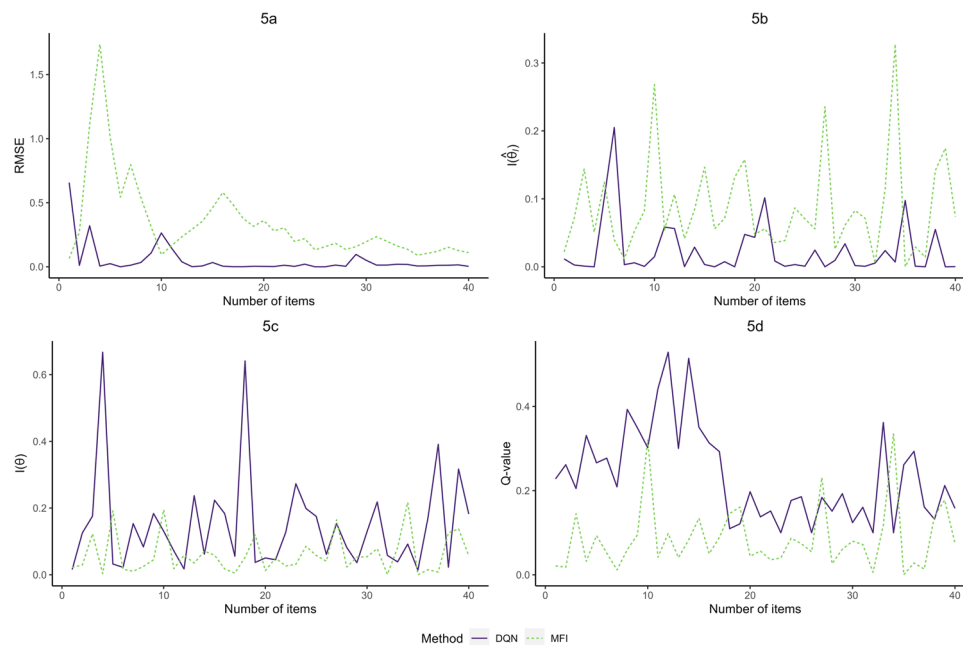


Fig. 5 RMSE, Fisher information calculated by estimated and true trait level, and Q-value of items selected by MFI and DQN for a specific examinee

Table 7 Descriptive statistics of the item parameters in the real item bank

Parameters	<i>M</i>	<i>SD</i>	Maximum	Minimum
<i>a</i>	1.045	0.379	0.486	2.351
<i>b</i>	0.383	1.751	−3.891	5.161
<i>c</i>	0.146	0.121	0.000	0.401

level (random sample from $[-0.5, 0.5]$), along with the ability estimation methods (Dodd's method and MLE), and the prior (standard normal distribution) and the estimation method (MAP for MEI) for the Bayesian item selection strategies, were in line with those used in the simulation study. Since one advantage of DQN lies in its ability to utilize the same network structure and hyper-parameters across similar problem scenarios (Mnih et al., 2015), the hyper-parameters and training process for DQN (details in Section "Implementation of the DQN algorithm") were kept the same as in the simulation.

Results

Figure 6 shows that the DQN strategy outperformed the other five strategies in terms of being unbiased under the real item bank and responses. In the initial five items of the test, the BIAS of the DQN strategy swiftly decreases to a level very close to 0 and then remains near 0 during the subsequent test. Traditional methods gradually reduce BIAS

as the test progresses, with only MLWI and MPWI having similar BIAS to DQN after 10 items.

Figure 7 shows that DQN had higher test accuracy and efficiency than the traditional strategies. The DQN strategy rapidly reduced the RMSE to a very low level within the first 10 items, a benchmark that traditional methods obtained after approximately 30 items. The DQN strategy then achieved a level of RMSE after selecting 20 items that traditional methods approximately reached after 40 items. Subsequently, the DQN consistently maintained the strategy with the lowest RMSE, which aligned with the results from the simulation study.

Discussion

In this paper, the reinforcement learning framework of the MDP was adopted in the context of CAT to calculate the long-run information of item selection, and the DQN method that could make full use of others' records was proposed to obtain the ideal item selection strategy. In addition, a detailed training algorithm was provided, and a simulation study was conducted to demonstrate the setup and processes used to obtain the best DQN models. The results of the two studies also showed that the new selection strategy based on DQN could achieve a lower test error on shorter tests and continuously improve measurement accuracy on longer tests for different kinds of item banks and groups of examinees; it demonstrated higher testing efficiency than traditional IRT-based strategies.

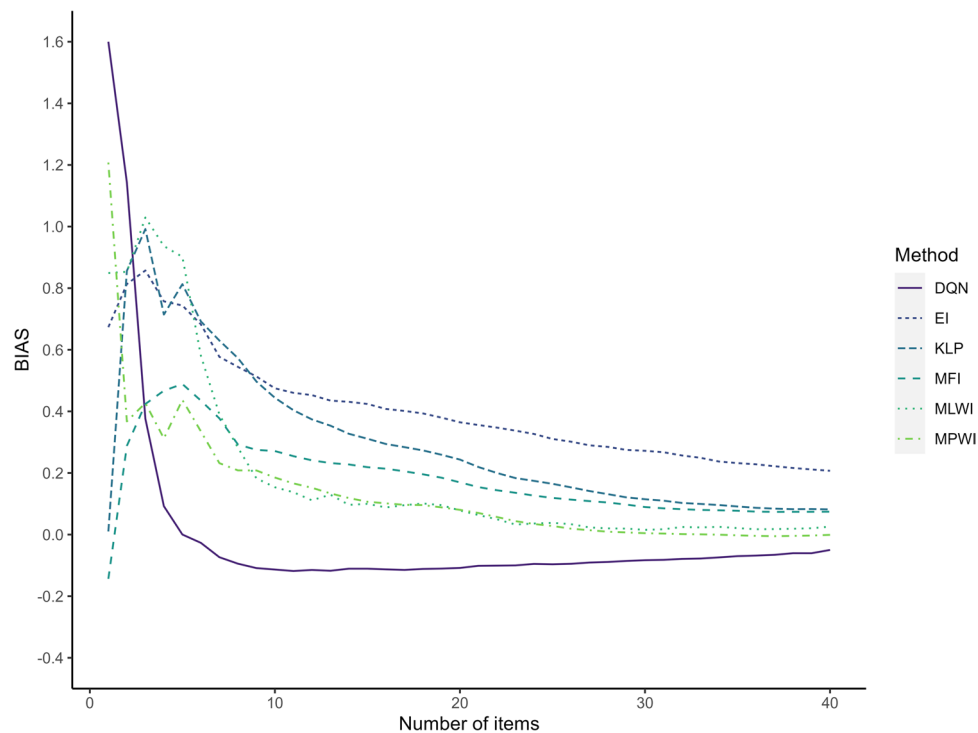


Fig. 6 BIAS of CAT employing DQN- and IRT-based strategies under the scenario in which the item bank and responses are real

Several factors are crucial when applying the DQN-based strategy in practice: the amount of data required to train the best Q-network, the distribution of abilities among the

examinees, and the time consumed in training and implementing the DQN for item selection. The number of training examinees needed to acquire the final best Q-network

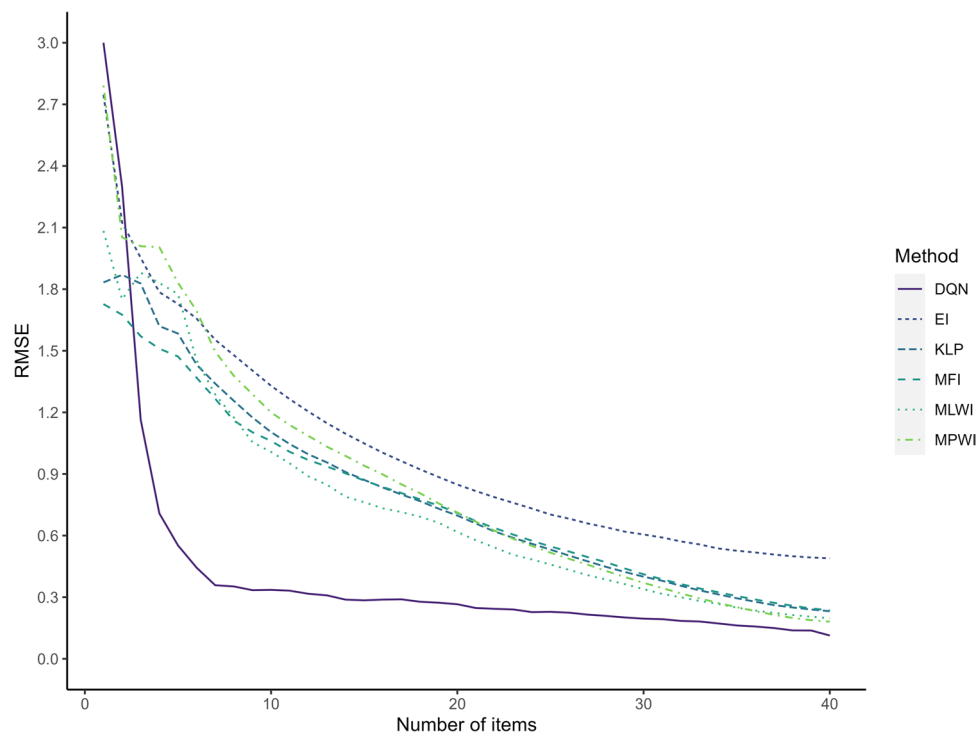


Fig. 7 RMSE of CAT employing DQN- and IRT-based strategies under the scenario where the item bank and responses are real

is 500 ± 300 , which was surprisingly small compared to N_{training} (1000, which is also the minimal sample size for the actor–critic algorithm used in CAT; see Nurakhmetov, 2019) and N_{testing} (5000, a common sample size in CAT studies; see, e.g., Barrada, Abad, & Veldkamp, 2009a, Barrada, Olea, et al., 2009b, 2010; Sorrel et al., 2020). This suggests that an efficient DQN-based selection strategy can be obtained with a small amount of data and can be generalized to a larger number of new examinees. These findings are aligned with previous research in adaptive learning, suggesting that the combination of DQN and a simulated environment model can produce excellent performance with a limited amount of data (Li et al., 2023).

Consistent with the principle of deep learning, the DQN trained by simulated responses generated by the standard normal distribution achieves higher accuracy than the traditional strategies for testing examinees with the same distribution of trait levels. It is worth noting that the DQN trained under uniform distribution is not affected by the inconsistent distribution of training and testing examinees but has higher test accuracy, especially for those with very high and very low trait levels. The reasons may be that the uniform distribution covers almost all possible trait levels of examinees in the training phase and that the training size is sufficiently large, which allows the DQN to fully interact with the examinees of each trait level and learn from their records. As a result, the overall performance of DQN with a uniform distribution is still better than that of the traditional strategies. Owing to this broad distribution, DQN significantly reduces the measurement error for examinees whose traits are at both ends of the distribution. In scenarios where the distribution of the testing data is uncertain, thereby precluding an exact match with the distribution of training data, this approach serves as a practical alternative.

In two studies, the training and employment of DQN to select each item is a speedy process. It takes only 200 seconds to conduct the 40-item test for 1000 examinees one by one to train the Q-network. In the validation and testing phases, it takes only 20 and 600 seconds for 200 and 5000

examinees to complete the 40-item test in parallel, respectively. If DQN is used for item selection, it takes less than one hundredth of a second on average to select an item for one examinee. The use of computational resources is also minimal. If Q-networks are trained on the GPU, the memory usage is 1000 MB, and the CPU utilization is 20% on average. If Q-networks are trained completely on the CPU, the speed and memory usage change slightly, and the CPU utilization only rises to 50%. This means that training can be performed equally quickly on a desktop or laptop without a GPU.

In the simulation study, we implicitly assumed that the 3PLM perfectly represented the examinees for generating responses, which is also important for a well-performing DQN-based strategy. Using the simulated response model has two main advantages: it reduces the time and economic costs when there is a lack of a sufficient amount of real response data with a wide range of trait levels, and it facilitates the exploration of various factors affecting the performance of DQN such as the distribution of trait levels and the number of examinees in the training phase. However, if the response model cannot accurately represent the examinees, the simulated responses and the trait levels based on them might be distorted, impairing the accurate estimation of the optimal strategy by DQN. Since the DQN algorithm directly estimates the Q-function without fitting an environmental model (Mnih et al., 2015), if sufficient and diverse real response data are available, the algorithm in Section "New item selection strategy based on the DQN algorithm" can be applied directly for training and testing DQN without the need to fit a student model to real data as the empirical study demonstrates. To summarize, based on the advantages and limitations of the DQN-based item selection strategy compared with traditional strategies, recommendations for different scenarios are outlined in Table 8.

This article is the first exploration of combining RL with CAT, and the feasibility of this strategy is proven. However, there remain practical issues worthy of discussion and innovation. First, the primary intention of the new selection

Table 8 Recommendations for using DQN-based and traditional IRT-based item selection strategies in different situations

	Use DQN-based strategy if	Use IRT-based strategy if
Desired outcome		
Trait accuracy	Emphasis on lower RMSE or MAE	Emphasis on unbiased estimation
Test objective	Aiming for higher test efficiency with accurate estimates in shorter tests	Aiming for precision along with other aspects such as exposure control and content balance
Implementation		
Data at hand	Substantial response data for all items or accurate response simulation across abilities	Limited response data for all items
Algorithm proficiency	Expertise in deep reinforcement learning	Familiarity with traditional methodologies

strategy proposed in this study is to increase the amount of test information, and satisfactory results were achieved on three indicators of the recovery of trait estimation. The problems of imperfect unbiased estimation, as well as other important targets of CAT such as exposure control and content balance, need further research. Second, elements of the MDP framework, such as the environment, the state, the reward, the MDP itself, and the DQN algorithm, can be adjusted or replaced with more advanced and implementable methods using RL or psychometric methods according to different types of CAT. We mainly followed the routines in IRT to define the reward and state. Future studies are encouraged so that better representations of these concepts can be explored and identified. For example, 3PLM and MLE were used for generating simulated records since the state is defined as a one-dimensional trait level. A variety of IRT and cognitive diagnostic models could also be adopted for multidimensional CAT and cognitive diagnostic CAT, which may warrant modification of the definitions of elements in the RL framework and the algorithm accordingly. In addition, the correct selection of models and coping with misfitting are also worthy of further study.

Conclusion

The Markov decision process framework of reinforcement learning is a powerful tool in item selection for computerized adaptive testing (CAT), as it corresponds to key concepts in CAT and IRT. This framework allows the definition of the Q-function as the expected item information to be obtained for all items in the bank at any given trait level, which can be calculated using DQN. The proposed strategy then selects the item with the highest expected accumulated weighted information. Two realistic studies demonstrated that the DQN-based item selection strategy outperforms traditional IRT-based methods for different item banks, test lengths, and examinees, with remarkable accuracy achieved using only a small number of simulated examinees and appropriate distribution settings to train the Q-networks. In conclusion, DQN provides a highly accurate and effective method for item selection in both simulated and real testing scenarios, outperforming traditional methods in terms of estimation accuracy.

Authors' contributions **Pujue Wang:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Hongyun Liu:** Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Mingqi Xu:** Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

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Data availability The datasets generated during the current study are available from the corresponding author upon reasonable request.

Code availability The code for all experiments is available on this project's Open Science Framework (OSF) page: <https://osf.io/nv3e4/>.

Declarations

Conflict of interest/Competing interests No authors report any financial or other conflicts of interest in relation to the work described.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

References

- Amit, R., Meir, R., & Ciosek, K. (2020, November 21). *Discount factor as a regularizer in reinforcement learning*. PMLR. <https://proceedings.mlr.press/v119/amit20a.html>
- Barrada, J. R., Abad, F. J., & Veldkamp, B. P. (2009a). Comparison of methods for controlling maximum exposure rates in computerized adaptive testing. *Psicothema*, 21(2), 313–320.
- Barrada, J. R., Olea, J., Ponsoda, V., & Abad, F. J. (2009b). Item selection rules in computerized adaptive testing: Accuracy and security. *Methodology*, 5(1), 7–17.
- Barrada, J. R., Olea, J., Ponsoda, V., & Abad, F. J. (2010). A method for the comparison of item selection rules in computerized adaptive testing. *Applied Psychological Measurement*, 34(6), 438–452.
- Chang, H. H., & Ying, Z. (1996). A global information approach to computerized adaptive testing. *Applied Psychological Measurement*, 20(3), 213–229.
- Chang, H. H., Qian, J., & Ying, Z. (2001). a-Stratified multistage computerized adaptive testing with b blocking. *Applied Psychological Measurement*, 25(4), 333–341.
- Chen, Y., Li, X., Liu, J., & Ying, Z. (2018). Recommendation system for adaptive learning. *Applied Psychological Measurement*, 42(1), 24–41.
- Chen, S. Y., Ankenmann, R. D., & Chang, H. H. (2000). A comparison of item selection rules at the early stages of computerized adaptive testing. *Applied Psychological Measurement*, 24(3), 241–255.
- Dodd, B. G. (1990). The effect of item selection procedure and step-size on computerized adaptive attitude measurement using the rating scale model. *Applied Psychological Measurement*, 14(4), 355–366.
- Frey, A. (2023). Computerized adaptive testing and multistage testing. In R. J. Tierney, F. Rizvi, & K. Erkican (Eds.), *International Encyclopedia of Education* (4th ed., pp. 209–216). Elsevier. <https://doi.org/10.1016/B978-0-12-818630-5.10028-4>
- Ghosh, A., & Lan, A. (2021, August 17). *BOBCAT: Bilevel optimization-based computerized adaptive testing*. arXiv.org. <https://arxiv.org/abs/2108.07386>
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT press.
- Gu, S., Holly, E., Lillicrap, T., & Levine, S. (2017). Deep reinforcement learning for robotic manipulation with asynchronous

- off-policy updates. *2017 IEEE international conference on robotics and automation (ICRA)* (pp. 3389–3396). IEEE.
- Han, R., Chen, K., & Tan, C. (2020). Curiosity-driven recommendation strategy for adaptive learning via deep reinforcement learning. *British Journal of Mathematical and Statistical Psychology*, 73(3), 522–540.
- He, W., Diao, Q., & Hauser, C. (2014). A comparison of four item-selection methods for severely constrained CATs. *Educational and Psychological Measurement*, 74(4), 677–696.
- Kingma, D. P., & Ba, J. (2014, December 22). *Adam: A method for stochastic optimization*. arXiv.org. <https://arxiv.org/abs/1412.6980>
- Li, X., Xu, H., Zhang, J., & Chang, H. H. (2021). Optimal hierarchical learning path design with reinforcement learning. *Applied psychological measurement*, 45(1), 54–70.
- Li, X., Xu, H., Zhang, J., & Chang, H. H. (2023). Deep reinforcement learning for adaptive learning systems. *Journal of Educational and Behavioral Statistics*, 48(2), 220–243.
- Lord, F. M. (1980). *Applications of item response theory to practical testing problems*. Erlbaum.
- Magis, D., & Barrada, J. R. (2017). Computerized adaptive testing with R: Recent updates of the package *catR*. *Journal of Statistical Software, Code Snippets*, 76(1), 1–18.
- Meijer, R. R., & Nering, M. L. (1999). Computerized adaptive testing: Overview and introduction. *Applied Psychological Measurement*, 23(3), 187–194.
- Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., & Riedmiller, M. (2013, December 19). *Playing atari with deep reinforcement learning*. arXiv.org. <https://arxiv.org/abs/1312.5602>
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., ..., & Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529–533.
- Nurakhmetov, D. (2019). Reinforcement learning applied to adaptive classification testing. In B. P. Veldkamp & C. Sluijter (Eds.), *Theoretical and Practical Advances in Computer-based Educational Measurement* (pp. 325–336). Springer. <https://doi.org/10.1007/978-3-030-18480-3>
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ..., & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. In *Advances in 33rd Conference on Neural Information Processing Systems* (pp. 8024–8035). MIT Press.
- Sallab, A. E., Abdou, M., Perot, E., & Yogamani, S. (2017). Deep reinforcement learning framework for autonomous driving. *Electronic Imaging*, 2017(19), 70–76.
- Samejima, F. (1977). A use of the information function in tailored testing. *Applied Psychological Measurement*, 1(2), 233–247.
- Schulman, J., Moritz, P., Levine, S., Jordan, M., & Abbeel, P. (2015, June 8). *High-Dimensional continuous control using generalized advantage estimation*. arXiv.org. <https://arxiv.org/abs/1506.02438>
- Shin, J., & Bulut, O. (2022). Building an intelligent recommendation system for personalized test scheduling in computerized assessments: A reinforcement learning approach. *Behavior Research Methods*, 54(1), 216–232.
- Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., Van Den Driessche, G., ..., & Hassabis, D. (2016). Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587), 484–489.
- Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., Guez, A., ..., & Hassabis, D. (2017). Mastering the game of go without human knowledge. *Nature*, 550(7676), 354–359.
- Sorrel, M. A., Barrada, J. R., de la Torre, J., & Abad, F. J. (2020). Adapting cognitive diagnosis computerized adaptive testing item selection rules to traditional item response theory. *Plos one*, 15(1), e0227196.
- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.
- Tan, C., Han, R., Ye, R., & Chen, K. (2020). Adaptive learning recommendation strategy based on deep Q-learning. *Applied psychological Measurement*, 44(4), 251–266.
- Tang, X., Chen, Y., Li, X., Liu, J., & Ying, Z. (2019). A reinforcement learning approach to personalized learning recommendation systems. *British Journal of Mathematical and Statistical Psychology*, 72(1), 108–135.
- van der Linden, W. J. (1998). Bayesian item selection criteria for adaptive testing. *Psychometrika*, 63(2), 201–216.
- van der Linden, W. J. (2016). *Handbook of Item Response Theory, Volume One: Models*. Chapman & Hall/CRC.
- van der Linden, W. J., & Reese, L. M. (1998). A model for optimal constrained adaptive testing. *Applied Psychological Measurement*, 22(3), 259–270.
- Veerkamp, W. J. J., & Berger, M. P. F. (1997). Some new item selection criteria for adaptive testing. *Journal of Educational and Behavioral Statistics*, 22(2), 203–226.
- Wainer, H., Dorans, N. J., Flaugher, R., Green, B. F., & Mislevy, R. J. (2000). *Computerized adaptive testing: A primer*. Routledge.
- Weiss, D. J. (1982). Improving measurement quality and efficiency with adaptive testing. *Applied Psychological Measurement*, 6(4), 473–492.
- Weiss, D. J., & Kingsbury, G. G. (1984). Application of computerized adaptive testing to educational problems. *Journal of Educational Measurement*, 21(4), 361–374.
- Wingersky, M. S., & Lord, F. M. (1984). An investigation of methods for reducing sampling error in certain IRT procedures. *Applied Psychological Measurement*, 8(3), 347–364.
- Zai, A., & Brown, B. (2020). *Deep reinforcement learning in action*. Manning Publications.
- Zhang, S., & Chang, H. H. (2016). From smart testing to smart learning: How testing technology can assist the new generation of education. *International Journal of Smart Technology and Learning*, 1(1), 67–92.

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